

THE PHYSICAL AI CO-DESIGN MATRIX						
Crossing Cognitive Agency (5 Lifecycle Rows) with Physical Reality (5 Constraint Columns)						
	1. TIME & FRESHNESS Δt_{wall} · P99 Tails	2. INERTIA & STOPPING $p = mv \cdot d_{stop}$ Kinematics	3. ACTUATION & JERK $\tau_e = L/R \cdot C^2$ Continuity	4. ENERGY & THERMAL $I^2 R$ Joule Heat · C_{th}	5. SILICON DETERMINISM UMA Bus QoS · <code>malloc=0</code>	
STAGE 1: PERCEIVE Vision · IMU · LiDAR Spatial Affordance Tokens → CHAPTER 4	Exposure & Sync Delay Sensor photon integration Global shutter trigger Hardware PTP timestamp Midpoint time registration	Dynamic Occlusion Moving obstacles obscure target Spatial clearance contraction Blind reaction zone expansion $d_{clearance} < d_{stop}$ risk	Transducer Noise IMU vibrational jitter Rolling shutter distortion Acoustic airframe resonance Encoder quantization error	Active Sensor Power LiDAR laser diode draw Camera sensor thermal rise ISP image pipeline energy Dark current noise drift	Zero-Copy DMA Ingestion MIPI CSI-2 bus serialization Lock-free ring buffers Zero userspace copy overhead Memory crossbar isolation	
STAGE 2: REMEMBER Temporal Belief Maps SE(3) Coordinate Trees → CHAPTER 5	Belief Age & TTL Information aging: $\Delta t = t - t_0$ Spatial covariance expansion $\sigma_{pos}(t) = \sigma_0 + v_{target} \cdot \Delta t$ Hard TTL lease expiration	Kinematic Propagation State prediction during dropout Dead reckoning with IMU Centrifugal frame tracking Momentum conservation model	Joint Compliance Drift Gearbox backlash deflection Cable stretch calibration Elastic linkage hysteresis SE(3) transform re-anchoring	SRAM Retention Power State memory power budget Latent cache leakage current Sleep mode state persistence Non-volatile state backup	Static Frame Trees Pre-allocated transform nodes Fixed-capacity spatial ring Zero dynamic heap allocation O(1) lookup determinism	
STAGE 3: REASON VLM Deliberation Expiring Intent Leases → CHAPTER 6	Asynchronous Leases Slow System 2 (0.5–2 Hz) Decoupled from motor loop Expiring intent lease L_{intent} Automatic timeout fallback	Kinematic Reachability Feasible workspace bounding Torque limit pre-screening Rejecting unachievable goals Centrifugal load filtering	Behavioral Smoothness Preventing goal thrashing Hysteresis in state switches Continuous affordance field Singularity avoidance	NPU Accelerator Power Edge accelerator TDP clamp Dynamic frequency scaling Thermal derating avoidance Energy per token budgeting	Mailbox Inter-Process Comms Shared SRAM atomic leases Monotonic sequence counters Hardware doorbell interrupts Lock-free multi-core ring	
STAGE 4: PLAN Diffusion Policies Action Chunking (ACT) → CHAPTER 7	Action Chunking Latency H-step future trajectories Amortizing inference time Temporal ensembling across H Masking bus jitter (20–50 Hz)	Dynamic Braking Horizon Planning stopping feasibility Reaction distance budget $d_{stop} = v \cdot \Delta t + v^2 / (2 \cdot a_{max})$ Safe corridor verification	C^2 Jerk Continuity Quintic polynomial splines Eliminating step-change torque Gearbox tooth shear protection Harmonic drive resonance bounds	Acceleration Energy Minimum-energy trajectories Peak motor current leveling Regenerative braking harvest Battery pack voltage sag clamp	Double-Buffered Chunks Ping-pong SRAM trajectory buffers Zero-contention slice access Atomic swap on chunk publish Deterministic MCU consumption	
STAGE 5: EXECUTE 1 kHz Safety Enforcer Control Barrier Functions → CHAPTER 8	Strict 1 kHz Loop Deadline 1000 Hz bare-metal execution Zero jitter: 1 ms ± 5 μs Hardware timer interrupt ISR Watchdog heartbeat refresh	CBF Safety Barrier Veto $h(x) \geq 0$ quadratic program solve Forward-invariant safe set Projects untrusted proposal u_t Emergency dynamic stop override	Inductive Voltage Limits $V(t) = RI + L \cdot di/dt + Ke \cdot \omega$ Field-Oriented Control (FOC) Back-EMF voltage saturation Gate-driver slew-rate control	I^2 t Continuous Current Stator thermal model tracking Winding insulation protection Inverter MOSFET over-temp Dynamic current derating	Sole PWM Authority Hardware timer peripheral lock Bus firewall blocks Linux MPU Safe Torque Off (STO) lines Static SRAM (malloc = 0)	